

| Question |     |       |    | Marking details                                                                                                                                                                                                                                                                                                                                | Marks available |          |          |          |          |          |
|----------|-----|-------|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |     |       |    |                                                                                                                                                                                                                                                                                                                                                | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
| 3        | (a) | (i)   |    | stimuli (1)<br>impulses (1)                                                                                                                                                                                                                                                                                                                    | 1<br>1          |          |          | 1<br>1   |          |          |
|          |     | (ii)  | I  | Neurones<br>Accept sensory neurone<br>Reject motor neurone/ relay neurone                                                                                                                                                                                                                                                                      | 1               |          |          | 1        |          |          |
|          |     |       | II | brain/spinal cord                                                                                                                                                                                                                                                                                                                              | 1               |          |          | 1        |          |          |
|          | (b) | (i)   |    | Any one (x1) from<br>• Hearing gets worse from 15/20 metres (for both pupils and teachers)<br>• {Teacher/ older people} hearing deteriorates more from 15/ 20 metres<br>• There is no change in hearing for 15 m/ up to 20m<br>• Everyone could hear (the buzzer) up to 15 m<br>• At {20/25}m more pupils than teacher could hear (the buzzer) |                 |          | 1        | 1        |          |          |
|          |     | (ii)  |    | Any one (x1) from: both groups (tested at):<br>same distances/<br>same buzzer/<br>same number of people                                                                                                                                                                                                                                        |                 | 1        |          | 1        |          | 1        |
|          |     | (iii) |    | Any one (x1) from:<br>length of time/<br>volume/loudness/<br>frequency of buzzer/<br>same age in each group/<br>same room/<br>same background noise/<br>same gender                                                                                                                                                                            |                 |          | 1        | 1        |          | 1        |
|          |     | (iv)  |    | more people (1)<br>more representative/ repeatability/ to spot anomalies (1)<br>Reject repeat experiment<br>Second mark point can only be awarded if first mark point correct                                                                                                                                                                  |                 |          | 2        | 2        |          | 2        |
|          |     |       |    | <b>Question 3 total</b>                                                                                                                                                                                                                                                                                                                        | <b>4</b>        | <b>1</b> | <b>4</b> | <b>9</b> | <b>0</b> | <b>4</b> |